Lecture 15-17 Scribe

L15_L17.pdf
PDF

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 15-17 slides). The target audience is a student preparing for a closed-notes exam.

Instructions:

Reason Step-by-Step: While writing the scribe, explicitly explain the reasoning behind definitions, assumptions, proofs, and example solutions exactly as presented in the provided context. Include this reasoning directly in the scribe text to help the student understand 'why' and 'how'.

Faithfulness: Include definitions, notation, theorem statements, and worked examples exactly as they appear in the slides. Do not hallucinate or add information from outside the context.

Logical Flow: Make the logical dependencies between concepts clear.

Structure: Organize the content with clear headings and subheadings corresponding to the lecture flow.

Goal: The output should allow a student to recall definitions, reconstruct proofs, and follow example solutions step-by-step."

Below is the **Lecture 15–17 Scribe (Joint Random Variables + Joint Distributions + Joint CDF)** written strictly from the slides.
All definitions, notation, steps, and examples are taken from the provided slides only.
Reasoning is included step-by-step so a student can **reconstruct derivations in a closed-notes exam**.
Source:

Lecture 15–17 Scribe

Joint Random Variables + Joint Distribution + Joint CDF

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1. Why Study Joint Random Variables?

In science and real life, we are often interested in **two (or more) random variables at the same time**.

Examples given in slides:

- Height and weight of giraffes
- IQ and birthweight of children
- Exercise frequency and heart disease rate
- Air pollution level and illness rate
- Number of Facebook friends and age
- Vehicle speed and traffic density
- Signal-to-noise ratio and throughput
- Rainfall and temperature
- Wind speed and delivery time
- Two dice experiment

Reasoning:
When two variables occur together, we need a way to compute probabilities involving both variables.

Slides state:

- Random variables have a **joint distribution**
- Joint distribution allows computing probability of events involving both variables
- If variables are independent → simplest case
- If not independent → we study relationship (covariance / correlation later)

So the goal of joint random variables is:

> Understand probability of events involving multiple variables and their relationship. (See slides 3–6)

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2. Discrete Case — Joint PMF

Setup

Suppose X and Y are discrete random variables.

$$X = \{x_1, x_2, \ldots, x_n\}$$

$$Y = \{y_1, y_2, \ldots, y_m\}$$

The ordered pair (X,Y) takes values in the product set

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

Reasoning:

Since both variables can take values, we must consider **all possible pairs**.

Definition (Joint PMF)

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Meaning:

This gives probability that both happen together.

> This gives probability of the joint outcome occurring simultaneously.
> (slide 8)

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3. Joint PMF Table Representation

Joint PMF can be written in table form.

X\Y	y1	y2	...	ym
x1	p(x1,y1)	p(x1,y2)	...	p(x1,ym)
x2	...	...	...	...
...	...	...	...	...
xn	...	...	...	p(xn,ym)

Key properties:

$$0 \leq p(x_i, y_j) \leq 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

Reasoning:

Total probability over all possible pairs must be 1.
(slide 10)

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4. Example — Rolling Two Dice

Experiment: roll two fair dice.

Define:

$$X = \text{value on first die}$$

$$T = \text{sum of both dice}$$

Each outcome (i,j) has probability

$$1/36$$

So we construct joint table of (X,T).

Observation from slide:

- Entries are 0 or 1/36
- X and T are not independent

Reasoning:

T depends on X → so not independent.
(slides 11–14)

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5. Continuous Case — Joint PDF

Continuous case is analogous to discrete case.

Slides state:

- Discrete values → continuous intervals
- PMF → PDF
- Sums → double integrals

If

$$X \in [a, b], \quad Y \in [c, d]$$

then

$$(X, Y) \in [a, b] \times [c, d]$$

Definition (Joint PDF)

A function f(x,y) is joint density if

$$P((X, Y) \text{ in small rectangle}) \approx f(x, y)dxdy$$

Reasoning:

Probability over area = density × area.
(slide 17–19)

Properties

$$f(x, y) \geq 0$$

$$\int_c^d \int_a^b f(x, y)dxdy = 1$$

Important note from slide:

> PDF may be greater than 1, because it is density not probability.

(slide 20)

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6. Continuous Example

Given:

$$f(x, y) = 4xy, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

Tasks:

1. Check valid PDF
2. Visualize event
3. Compute probability

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

(slide 21–22)

Step 1 — Check Validity

Check normalization:

$$\int_0^1 \int_0^1 4xydxdy$$

First integrate w.r.t x:

$$\int_0^1 4y \left[\frac{x^2}{2}\right]_0^1 dy = \int_0^1 2ydy$$

Now integrate:

$$\int_0^1 2ydy = [y^2]_0^1 = 1$$

So valid PDF.

Reasoning:

Total probability must be 1.
(slide 24–25)

Step 2 — Compute P(A)

Event:

$$X < 0.5, Y > 0.5$$

Limits:

$$0 \leq x \leq 0.5$$

$$0.5 \leq y \leq 1$$

Probability:

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy\, dydx$$

Integrate:

$$\int_0^{0.5} 2x(1 - 0.25)dx$$

$$= \int_0^{0.5} \frac{3}{2} x dx$$

$$= \frac{3}{16}$$

(slide 26)

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7. Joint CDF

Notation:

$$X \leq x, \quad Y \leq y$$

means event

$$\{X \leq x \text{ and } Y \leq y\}$$

Definition:

$$F(x, y) = P(X \leq x, Y \leq y)$$

(slide 29)

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8. Joint CDF — Continuous Case

If joint PDF is f(x,y)

then

$$F(x, y) = \int_c^y \int_a^x f(u, v) du dv$$

Reasoning:

CDF = probability up to x and y → integrate density.

Recover PDF:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

(slide 32–33)

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9. Joint CDF — Discrete Case

If PMF = p(xi, yj)

then

$$F(x, y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

Interpretation:

- Add probabilities in table
- rows with xi ≤ x
- columns with yj ≤ y

(slide 35–37)

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10. Properties of Joint CDF

1.

$$F(x, y) \text{ is non-decreasing}$$

If x or y increases → F increases.

2.

$$F(x, y) = 0$$

at lower-left corner.

3.

$$F(x, y) = 1$$

at upper-right corner.

Reasoning:

CDF accumulates probability from −∞ to point.

(slides 39–41)

Final Summary (Exam Ready)

You must remember:

- Joint PMF definition

- Joint PDF definition
- Normalization condition
- Table representation
- Continuous double integral
- Joint CDF definition
- CDF from PDF
- PDF from CDF
- Discrete CDF summation
- Properties of CDF
- Example with 4xy

These are the main exam questions.

✅ If you want, I can also make **formula sheet for L15-17 (1 page revision)**.

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Below is the **fully faithful, step-by-step lecture scribe for Lecture 15–17**, written strictly from the slides, with reasoning included inside the text so a student can reconstruct definitions, proofs, and examples in a closed-notes exam.
All content comes from the provided slides.
Source:

Lecture 15–17 Scribe

Joint Random Variables + Joint Distribution + Joint PDF + Joint CDF

---

1. Why Study Joint Random Variables?

In real life we are often interested in **two (or more) random variables at the same time**.

Examples given in slides:

- Height and weight of giraffes
- IQ and birthweight of children
- Exercise frequency and heart disease rate
- Air pollution level and respiratory illness
- Number of Facebook friends and age
- Vehicle speed and traffic density
- SNR and throughput in wireless network
- Rainfall and temperature
- Wind speed and delivery time
- Rolling two dice

Reasoning

When two variables occur together, we must compute probabilities involving both variables.

Slides state:

- Random variables have a **joint distribution**
- Joint distribution allows probability of events involving both variables
- If variables are independent → simplest case
- If not independent → need joint analysis

Therefore,

> We study joint random variables to understand the relationship between variables and compute probabilities involving both.

---

2. Discrete Case — Joint PMF

2.1 Setup

Suppose X and Y are discrete random variables.

$$X = \{x_1, x_2, \ldots, x_n\}$$

$$Y = \{y_1, y_2, \ldots, y_m\}$$

Then the ordered pair (X,Y) takes values in

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

Reasoning

Since both variables can vary, we must consider all possible pairs of values.

This set is called the **product set**.

2.2 Definition — Joint PMF

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Meaning:

Probability that both events happen together.

Reasoning

Single variable → P(X=x)

Two variables → P(X=x AND Y=y)

So we need joint probability.

Slides state:

> This gives probability of the joint outcome occurring simultaneously.

2.3 Joint PMF Table Representation

Joint PMF is written in table form.

X\Y	y1	y2	...	ym
x1	p(x1,y1)	p(x1,y2)	...	p(x1,ym)
x2	...	...	...	...
...	...	...	...	...
xn	...	...	...	p(xn,ym)

Properties

$$0 \leq p(x_i, y_j) \leq 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

Reasoning

All probabilities must be between 0 and 1

Total probability of all outcomes must be 1.

---

3. Example — Rolling Two Dice

Experiment:

Roll two fair dice.

Define

$$X = \text{value on first die}$$

$$T = \text{sum of both dice}$$

Each outcome (i,j) has probability

$1/36$

So we build joint table of (X,T).

Observation from slide:

- Entries are 0 or 1/36
- X and T are not independent

Reasoning

T depends on X

Example:

If X=6 → T must be ≥6

So not independent.

---

4. Continuous Case — Joint PDF

Discrete → continuous analogy:

Discrete	Continuous
PMF	PDF
Sum	Integral
Points	Intervals

If

$$X \in [a, b], \quad Y \in [c, d]$$

then

$$(X, Y) \in [a, b] \times [c, d]$$

Reasoning

Two variables → region in plane.

4.1 Definition — Joint PDF

Function f(x,y) is joint density if

$$P((X,Y) \text{ in small rectangle}) \approx f(x,y)dxdy$$

Reasoning

Probability = density × area

Because continuous variables use area instead of points.

4.2 Properties

$$f(x,y) \geq 0$$

$$\int_c^d \int_a^b f(x,y)dxdy = 1$$

Important note from slide:

> f(x,y) may be greater than 1
> because it is density, not probability.

---

5. Example — Joint PDF

Given

$$f(x,y) = 4xy$$

for

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

Tasks:

1. Check valid PDF
2. Visualize event
3. Compute probability

Event

$$A = \{X < 0.5, Y > 0.5\}$$

5.1 Step 1 — Check Validity (Normalization)

We check

$$\int_0^1 \int_0^1 4xydxdy$$

Integrate w.r.t x

$$\int_0^1 4y \left[\frac{x^2}{2}\right]_0^1 dy = \int_0^1 2ydy$$

Integrate

$$\int_0^1 2ydy = [y^2]_0^1 = 1$$

Hence valid PDF.

Reason:

Total probability must equal 1.

5.2 Step 2 — Compute P(A)

Event:

$$0 \leq x \leq 0.5$$

$$0.5 \leq y \leq 1$$

Probability

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xydydx$$

Integrate y

$$= \int_0^{0.5} 2x(1 - 0.25)dx$$

$$= \int_0^{0.5} \frac{3}{2}xdx$$

$$= \frac{3}{16}$$

---

6. Joint Cumulative Distribution Function

Notation

$$X \leq x, Y \leq y$$

means

$$\{X \leq x \text{ and } Y \leq y\}$$

Definition

$$F(x, y) = P(X \leq x, Y \leq y)$$

Reasoning

CDF = probability up to a point.

---

7. Joint CDF — Continuous Case

If PDF = f(x,y)

then

$$F(x, y) = \int_c^y \int_a^x f(u, v) du dv$$

Reason:

CDF accumulates probability.

Recover PDF

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

Reason:

Derivative of CDF gives density.

---

8. Joint CDF — Discrete Case

If PMF = p(xi,yj)

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

Interpretation:

Add probabilities

- rows ≤ x
- columns ≤ y

Reason:

CDF adds all probabilities up to point.

---

9. Properties of Joint CDF

1.

$$F(x, y) \text{ is non-decreasing}$$

If x or y increases → F increases.

Reason:

More region → more probability.

2.

$$F(x, y) = 0$$

at lower-left corner.

Reason:

No probability yet.

3.

$$F(x, y) = 1$$

at upper-right corner.

Reason:

All probability included.

Final Exam Summary

Must remember:

- Joint PMF definition
- Joint PMF table

- Sum = 1 property
- Joint PDF definition
- Double integral = 1
- Continuous example 4xy
- Joint CDF definition
- CDF from PDF
- PDF from CDF
- Discrete CDF sum
- CDF properties
- Dice example

These are the main exam questions from L15-17.

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definitions, reconstruct proofs, and follow example solutions step-by-step."

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Lecture 15–17 Scribe

Joint Random Variables • Joint PMF • Joint PDF • Joint CDF

---

1. Why Study Joint Random Variables?

In science and real life, we are often interested in **two (or more) random variables at the same time**.

Examples given in slides:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and heart disease rate
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age
- Vehicle speed and traffic density
- Signal-to-Noise Ratio and throughput
- Rainfall and temperature
- Wind speed and delivery time
- Rolling two dice

Reasoning

When two quantities vary together, we must compute probability involving both variables.

Slides state:

- Random variables have a **joint distribution**
- Joint distribution allows computing probability of events involving both variables

- If variables are independent → simplest case
- If not independent → need joint analysis

Therefore,

> Joint random variables are needed to study the relationship between variables and compute probabilities involving both.

---

2. Discrete Case — Joint PMF

2.1 Setup

Suppose X and Y are discrete random variables.

$$X = \{x_1, x_2, \ldots, x_n\}$$

$$Y = \{y_1, y_2, \ldots, y_m\}$$

Then the ordered pair (X,Y) takes values in

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

Reasoning

Since X can take n values and Y can take m values,
all possible outcomes are pairs → product set.

2.2 Definition — Joint PMF

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

Meaning:

Probability that X = xi and Y = yj simultaneously.

Slides state:

> This gives the probability of the joint outcome occurring simultaneously.

Reasoning

Single variable → P(X=x)

Two variables → P(X=x AND Y=y)

So we define joint PMF.

2.3 Joint PMF Table Representation

Joint PMF is organized in a table.

X\Y	y1	y2	...	ym
x1	p(x1,y1)	p(x1,y2)	...	p(x1,ym)
x2	...	...	...	...
...	...	...	...	...
xn	...	...	...	p(xn,ym)

Properties

$$0 \leq p(x_i, y_j) \leq 1$$

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

Reasoning

All probabilities must be non-negative

Total probability of all possible pairs must be 1.

---

3. Example — Rolling Two Dice

Experiment:

Roll two fair dice.

Define

$$X = \text{value on first die}$$

$$T = \text{sum of both dice}$$

Each outcome (i,j) has probability

$$1/36$$

So we construct joint probability table of (X,T).

Observation from slide:

- Entries are either 0 or 1/36
- X and T are not independent

Reasoning

T depends on X.

Example:

If X = 6 → T must be ≥ 6

So knowing X changes probability of T.

Therefore not independent.

---

4. Continuous Case — Joint PDF

Slides say continuous case is analogous to discrete case.

Changes:

Discrete	Continuous
Values	Intervals
PMF	PDF
Sum	Double integral

If

$$X \in [a, b], \quad Y \in [c, d]$$

then

$$(X, Y) \in [a, b] \times [c, d]$$

Reasoning

Two variables → region in plane.

4.1 Definition — Joint PDF

A function f(x,y) is joint density if

$$P((X,Y) \text{ lies in small rectangle}) \approx f(x,y)dxdy$$

Reasoning

For continuous variables:

Probability of a point = 0

Probability over area = density × area

4.2 Properties of Joint PDF

$$f(x,y) \geq 0$$

$$\int_c^d \int_a^b f(x,y)dxdy = 1$$

Slide note:

> Joint PDF may be greater than 1
> because it is density, not probability.

Reasoning

Total probability must be 1
Density can be large if region small.

---

5. Example — Joint PDF

Given

$$f(x,y) = 4xy$$

for

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

Tasks:

1. Show valid PDF
2. Visualize event
3. Compute probability

Event

$$A = \{X < 0.5, Y > 0.5\}$$

5.1 Step 1 — Check Validity

Check normalization

$$\int_0^1 \int_0^1 4xydxdy$$

Integrate w.r.t x

$$\int_0^1 4y \left[\frac{x^2}{2}\right]_0^1 dy = \int_0^1 2ydy$$

Integrate

$$\int_0^1 2ydy = [y^2]_0^1 = 1$$

Hence

- f ≥ 0
- integral = 1

So valid joint PDF.

Reason:

Total probability must equal 1.

5.2 Step 2 — Compute P(A)

Event

$$X < 0.5, Y > 0.5$$

Limits

$$0 \leq x \leq 0.5$$

$$0.5 \leq y \leq 1$$

Probability

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xydydx$$

Integrate y

$$= \int_0^{0.5} 2x(1-0.25)dx$$

$$= \int_0^{0.5} \frac{3}{2} x dx$$

$$= \frac{3}{16}$$

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6. Joint Cumulative Distribution Function

Notation

$$X \leq x, \quad Y \leq y$$

means

$$\{X \leq x \text{ and } Y \leq y\}$$

Definition

$$F(x,y) = P(X \leq x, Y \leq y)$$

Reasoning

CDF gives probability accumulated up to (x,y).

---

7. Joint CDF — Continuous Case

If PDF = f(x,y)

then

$$F(x,y) = \int_c^y \int_a^x f(u,v) du dv$$

Reason:

CDF = integral of density.

Recover PDF

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

Reason:

Derivative of CDF gives density.

---

8. Joint CDF — Discrete Case

If PMF = p(xi,yj)

$$F(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

Interpretation from slide:

- Add probabilities in table
- rows with xi ≤ x
- columns with yj ≤ y

Reason:

CDF accumulates probability.

---

9. Properties of Joint CDF

1.

$$F(x,y) \text{ is non-decreasing}$$

If x or y increases → F increases.

Reason:

More region included.

2.

$$F(x,y) = 0$$

at lower-left of range.

Reason:

No probability yet.

3.

$$F(x, y) = 1$$

at upper-right of range.

Reason:

All probability included.

Final Exam Recall List

Must remember:

- Joint PMF definition
- Joint PMF table
- Sum = 1 property
- Dice example
- Joint PDF definition
- Double integral = 1
- 4xy example
- Joint CDF definition
- Continuous CDF integral
- Discrete CDF sum
- PDF from CDF derivative
- CDF properties

These are exactly the concepts in Lecture 15–17.